

ACRME — Technical Design Document (TDD)

Title	Azure Capacity Reservation Management Engine (ACRME) — Technical Design Document
Version	1.0 (net-new)
Date	2 September 2026
Status	Draft for review — supersedes the Production Readiness Review as the technical design of record
Baseline	Azure Capacity & Quota Management — Consolidated Requirements Baseline v2.2 (27 Aug 2026)
Owner	Vishnuvardhan Reddy · Principal Cloud Architect
Audience	Engineering, SRE/platform operations, security, POC leads
Companion	Functional Design Document (<code>acrme_functional_design_document.md</code>)
Decision records	ADR-001 (region selection), ADR-002 (quota & capacity, single-pool), ADR-003 (capacity during DR), ADR-004 (forecast & increase), ADR-005 (distributed DR reference model) — all v2.2

Purpose. This TDD describes **how** ACRME is built — components, runtime, topology, data, state, algorithms, interfaces, security, observability, NFRs, integration, and POC gating — traceable to Requirements Baseline v2.2 and to the companion FDD (which owns the what). This document is **self-contained**: all normative algorithms, formulas, schemas, enumerations, and constants are inlined, not referenced externally.

Reconciliation note (v2.2). This TDD implements the confirmed v2.2 decisions: **single governed quota pool** as the primary technical model with logical earmarks for Prod/DR protection (QUA-004, ADR-002); **Switzerland North** as the EU cross-geo DR extension region (REG-002) — pre-configured but **conditional and currently inactive** for the Middle East, whose current legal position is `DR_NOT_OFFERED` pending DEC-001 (see §2.2 constraints); **max-not-sum** destination DR sizing with an authoritative `SourceDestinationDRIndex` (DR-016/017/018, ADR-005); **exact-production-region-first** validation with a governed `CustomerSeedRecord` (PLC-001..005, ADR-001); **five-state engine machine** and **standby activation waves** (DR-019, ADR-003).

1. Introduction

1.1 Scope

This document specifies the technical design of ACRME: the control engine that guarantees every managed Azure deployment has **both** reserved physical capacity and deployable VM-family quota — in the correct region/zone/SKU — before it proceeds, while minimising idle cost through lean distributed DR and continuous reconciliation. It covers component architecture, runtime and deployment, data and state, the placement/scoring/DR algorithms, the API surface, security, observability, non-functional behaviour, integration, and the POC dependencies that gate production reliance.

1.2 Relationship to the FDD

The FDD (`acrme_functional_design_document.md`) defines **what** ACRME does — capabilities, flows, states, and rules — traceable to Baseline v2.2. This TDD defines **how** those capabilities are realised. Where the FDD names a behaviour (e.g. “single governed pool”, “max-not-sum DR”, “readiness states”), this TDD gives the component, schema, algorithm, and interface that implements it. Section references to the FDD use the form (FDD §4.x).

1.3 Decision records referenced

ADR	Title	Key technical mandate used here
ADR-001	Region selection & customer placement	Exact-region-first validation; seed record; staged placement pipeline; input modes
ADR-002	Quota & capacity management	Single governed pool (primary) ; logical earmarks; exception two-group topology
ADR-003	Capacity management during DR	Five-state engine machine; three-tier emergency transfer; quota-neutral Tier 2
ADR-004	Forecast & increase of capacity/quota	Reservation floor vs forecast growth; auto-increase triggers; 10-step lifecycle
ADR-005	Distributed DR reference model	<code>SourceDestinationDRIndex</code> ; max-not-sum; overcommit ratio/gap; §12A worked model

1.4 Evidence tags

Every non-trivial design assertion carries an evidence tag consistent with the requirements corpus: `[Documented]` (Microsoft Learn / Azure platform), `[Decided]` (approved ADR decision), `[Derived]` (logical consequence; POC where noted), `[Assumed]` (tunable policy default, not yet empirically validated).

2. Architecture Overview

2.1 Design principles (baseline §20)

1. **Readiness is capacity $\wedge$ quota $\wedge$ freshness.** No deployment is “ready” on a reservation alone; deployable consumer quota and non-stale state are equally required (RDY-003, NFR-002). [Decided]
2. **Continuous reconciliation of desired vs actual.** A reservation floor (`Allocated + Buffer` , CAP-003) is enforced every cycle; forecast growth is a separate proactive path (ADR-004). [Decided]
3. **Lean, distributed, reciprocal DR.** No dedicated DR regions; each region hosts Prod + CVAL + DR standby for other regions; standby is sized max-not-sum and shared across mutually exclusive failures (DR-016/017, ADR-005). [Decided]
4. **Single governed quota pool.** Collect all eligible per-region quota into one pool per region/quota family and allocate on demand; protect Prod/DR by logical earmark, minimising quota-increase requests to Microsoft (QUA-004, ADR-002). [Decided]
5. **Config-as-code, deterministic, replayable.** All policy (weights, buffers, thresholds, catalogue) is versioned; every decision records inputs, policy version, and snapshot ref for deterministic replay. [Decided]
6. **Fail-safe and idempotent.** Stale state blocks placement; all mutations are idempotent with operation handles and optimistic concurrency (NFR-002/007). [Decided]

2.2 Constraints (baseline §21)

- **Middle East DR is `DR_NOT_OFFERED` (current legal position, DEC-001 pending).** The Middle East programme is legal-owned and serves government/medical customers under data-residency/sovereignty laws; cross-border DR cannot meet residency requirements, so **no DR is offered there today** (baseline §2, §6, DR-014, DEC-001). Middle East **production** may still be placed and governed; the placement/DR engine emits `dr_region = NOT_OFFERED` for those geographies and creates **no** `SourceDestinationDRIndex` entry, **no** DR earmark, and **no** cross-geo activation. The Switzerland North cross-geo path (REG-002/HC-10) is **pre-configured but inactive** for the Middle East; it activates **only** if Legal clears DEC-001, via a config flip (`dr_not_offered["Middle East"] = false`) with **no code change**. `DR_NOT_OFFERED` is evaluated **before** any cross-geo extension. [Documented]
 - **Azure Quota Group / `groupQuotas` maturity** gates single-pool enforcement (DEP-001); the two-group topology is the sanctioned fallback. [Documented]
 - **No native intra-pool sub-reservation** in Azure quota groups — the DR earmark and Prod floor are engine-enforced arithmetic, not platform primitives (Scenario 9). [Documented]
 - **ARM throttling budgets** bound reconciliation cadence and estate size (Compute RP: 250 reads / 5 min, 1,200 writes / hour per subscription). [Documented]
 - **Phase 1 guardrails:** destructive VM-association transfers (Tier 3) are blocked; auto-decrease is operator-gated; forecast is advisory. [Decided]
 - **Consumer-quota-under-shared-reservation (POC-001)** is the top unknown; readiness must validate consumer quota until proven. [Assumed]
-

3. Logical Component Architecture (§23)

ACRME is a set of cooperating services around a versioned state store and a policy (config-as-code) service. Components are deployable as modules of a single control application; boundaries are drawn for testability and independent scaling.

Component	Responsibility	Key requirements
Inventory Collector	Polls Azure (CRGs, reservations, quota/groupQuotas, usage, associations, zones) into freshness-stamped snapshots.	DAT-001..004, CAP-001/002, NFR-002
State Reconciler	Every cycle compares desired vs actual; enforces Allocated + Buffer floor; guarded scale-down; never deletes.	CAP-003/004/009/010, OPS-002
Placement & Scoring Engine	Validates exact region (or derives on exception); computes PS_Prod/PS_NonProd/PS_DR ; writes seed record.	PLC-001..010, REG-001..003
Quota-Pool Manager	Maintains the single governed pool; computes earmarks; allocates on demand; enforces Prod floor & DR earmark; exception two-group mode.	QUA-001..014, ADR-002
DR Orchestrator	Maintains SourceDestinationDRIndex ; sizes max-not-sum; runs standby activation waves & staged acquisition; fallback.	DR-001..019, ADR-005
Readiness API	Combined capacity+quota+freshness verdict as a machine-readable state to AEP.	RDY-001..004, INT-001..007
State Store	Versioned, optimistic-concurrency document store for entities, snapshots, operation/audit records.	DAT-001..006, NFR-007
Config / Scope-File Service	Versioned PlacementPolicy , region catalogue, scope files; decision-traceable.	GOV-003, DAT-005, REG-001
Forecast & Increase Service	Advisory Forecast_Quantity ; auto-increase triggers; oper-	CAP-005, ADR-004

Component	Responsibility	Key requirements
	ator-gated capacity/quota increase.	
Emergency Transfer Service	Tier 1/2/3 escalation during DR_EVENT_ACTIVE ; quota-neutral Tier 2; Tier 3 blocked in Phase 1.	ADR-003, §13 baseline
Observability Emitter	Metrics, alerts, dashboards incl. per-destination max-source coverage & source↔destination map.	OBS-001..005
Governance & Audit	RBAC, managed identity, approval gates, break-glass, immutable audit with before/after + correlation IDs.	GOV-001..009

T1 — Logical component architecture

```

flowchart TB
    subgraph External
        AEP[AEP / Provisioning]
        OPS[Operators / DR coordinator]
        AZ[Azure control planes:<br/>Compute / Reservations / groupQuotas]
        OBS[Observability platform]
    end
    subgraph ACRME
        API[Readiness API]
        PE[Placement & Scoring Engine]
        REC[State Reconciler]
        QPM[Quota-Pool Manager]
        DRO[DR Orchestrator]
        FIS[Forecast & Increase Service]
        ETS[Emergency Transfer Service]
        INV[Inventory Collector]
        CFG[Config / Scope-File Service]
        GOV[Governance & Audit]
        OE[Observability Emitter]
        STORE[(State Store<br/>versioned + optimistic concurrency)]
    end
    AEP -->|readiness query / mutations| API
    OPS -->|approvals, DR declaration| API
    API --> PE & QPM & DRO & REC
    PE --> STORE
    QPM --> STORE
    DRO --> STORE
    REC --> STORE
    FIS --> STORE
    ETS --> STORE
    INV -->|snapshots| STORE
    INV <--> AZ
    QPM <--> AZ
    REC <--> AZ
    DRO <--> AZ
    CFG --> PE & QPM & DRO & REC & FIS
    GOV -.enforces.-> API
    OE --> OBS
    STORE --> OE

```

4. Runtime & Deployment (§23, CAP-006, PLC-009)

4.1 Execution model

- **Reconciliation job (container app):** a long-running scheduler executes the reconciliation loop on a **6-minute target interval (configurable, CAP-006)**; targeted/critical reconciles run out-of-band. It drives the Inventory Collector → State Reconciler → Quota-Pool Manager cycle and emits snapshots. [Decided]
- **Placement flow (function/event path, PLC-009):** onboarding and AEP readiness requests are handled synchronously against the latest snapshot; on stale state the request triggers a synchronous refresh or returns `STALE_STATE`. [Decided]
- **DR path:** DR declaration flips `engine_mode` to `DR_EVENT_ACTIVE`; the DR Orchestrator drives priority-wave activation out of the steady-state cadence. [Decided]

4.2 Cadence, SLOs and drift (§38, NFR)

```

Reconciliation loop target      = 6 min (configurable; production interval TBD)
[Assumed]
Critical targeted reconcile P95 < 2 min
[Assumed]
Stable-resource state age P95 < 15 min
[Assumed]
Drift detection                  ≤ 2 reconciliation cycles
[Assumed]
Snapshot max age (staleness gate) = max_snapshot_age_seconds (config, RDY-004)
[Decided]

```

4.3 API surface (summary; detail in §11)

A readiness/placement API (query + idempotent mutations), an operations/approval API (increase approval, DR declaration, override, sharing activation), and a read-only audit/observability API. All mutating operations are idempotent and versioned (INT-004..006). [Decided]

T2 — Management group / subscription topology

```

flowchart TB
    MG[Management Group root] --> MGPlat[Platform MG]
    MG --> MGWork[Workload MGs]
    MGPlat --> SubCtl[ACRME control subscription<br/>engine, state store, config]
    MGPlat --> SubObs[Observability subscription]
    MGWork --> SubProd[Prod subscriptions<br/>Prod CRGs + pool]
    MGWork --> SubNonProd[NonProd/CVAL subscriptions<br/>NonProd CRGs]
    MGWork --> SubDR[DR-hosting subscriptions<br/>DR standby CRGs]
    SubCtl -. managed identity + RBAC .-> SubProd
    SubCtl -. managed identity + RBAC .-> SubNonProd
    SubCtl -. managed identity + RBAC .-> SubDR

```

5. Topology — Subscriptions, CRGs and Quota Groups (§24, §25, §26)

5.1 Management group / subscription model (§24)

ACRME runs from a platform control subscription with cross-subscription managed-identity access (least-privilege RBAC) to the Prod, NonProd/CVAL, and DR-hosting subscriptions (see T2). All mutations are attributed to the control identity and audited (GOV-002). [Decided]

5.2 CRG hierarchy and cross-subscription sharing (§25, FC-06)

- **Three-CRG model per region:** exactly one Prod, one NonProd/CVAL, and one DR-standby CRG target per region per scope (CAP-001). [Decided]
- **Zone alignment (FC-06):** cross-subscription CRG sharing requires a validated `ZoneMappingRecord` — logical zones differ across subscriptions, so sharing is only valid when physical zones align (CAP-014..019). [Decided]
- **Sharing limits:** provider→consumer sharing validated for SKU/region/zone/authorisation; a per-provider consumer ceiling applies. [Decided]

T3 — CRG hierarchy & cross-subscription sharing (zone alignment)

```

flowchart TB
    subgraph RegionR [Region R]
        ProdCRG[Prod CRG]
        NonProdCRG[NonProd/CVAL CRG]
        DRCRG[DR-standby CRG]
    end
    ProdCRG -->|reserves| ZmapP[{{Zone map: logical->physical}}]
    NonProdCRG --> ZmapN[{{Zone map}}]
    DRCRG --> ZmapD[{{Zone map}}]
    ZmapP --> ShareA[Consumer subscription A<br/>zone-aligned share]
    ZmapN --> ShareB[Consumer subscription B]
    ZmapD --> ShareC[DR consumer subscription]
    note1[FC-06: share valid only when physical zones align<br/>ZoneMappingRecord required]:::n
    classDef n fill:#fff3cd,stroke:#e0a800;

```

5.3 Quota-group architecture — single-pool decision (§26, QUA-004)

Primary model: one governed quota group per region and quota family, covering Prod + NonProd/CVAL + DR together. All eligible default per-region quota is collected into the pool (QUA-003) and allocated on demand. Prod and DR are protected inside the shared pool by **logical earmarks** enforced by the Quota-Pool Manager at allocation/reclamation time — not by physical group separation. Because all three environments share one pool, an emergency DR draw needs **no cross-group transfer**. [Decided]

Exception topology: two (or more) groups per region, used **only** when Azure Quota Group limits or a mandatory Prod-isolation governance boundary make one pool impossible; each case is recorded in the Decision Log and reverts to one pool when the constraint lifts (DEP-001). [Decided]

The pool arithmetic (`Pool_Limit` , `Prod_Reserved_Floor` , `DR_Earmark_vCPU` , `Allocatable_NonProd` , `Pool_Headroom`) is specified in §8.3. See T4.

T4 — Quota-group / pool architecture (single-pool model)

```

flowchart TB
    subgraph Pool [Single governed quota pool per region/quota family]
        direction TB
        PF[Prod_Reserved_Floor<br/>= Prod_Used + Prod_Growth_Buffer<br/>never allocatable to NonProd]
        DE[DR_Earmark_vCPU<br/>= Destination_DR_Requirement x vCPU max-not-sum<br/>never allocatable to NonProd]
        AN[Allocatable_NonProd<br/>= Pool_Limit - Prod_Reserved_Floor - DR_Earmark - NonProd_Used]
        PH[Pool_Headroom = Pool_Limit - Pool_Used]
    end
    Pool -->|emergency DR draw: pool headroom first,<br/>then reclaimable NonProd - no cross-group transfer| DR[DR activation]
    Exc[Exception topology only:<br/>Prod group + NonProd+DR group<br/>when Azure limit / Prod-isolation boundary]:::e
    classDef e fill:#f8d7da,stroke:#dc3545;

```

6. Data Architecture (§34, DAT-001..006)

6.1 Store characteristics

A versioned document store with per-document optimistic concurrency (`policy_version` / `_etag` -style guards), freshness metadata on every snapshot, and append-only operation/audit records (DAT-001..006, NFR-007). Every decision-driving read carries a `capacity_snapshot_ref` for deterministic replay. [Decided]

6.2 Core entities

The three v2.2-critical entities are `CustomerSeedRecord`, `SourceDestinationDRIndex`, and `CVALEarmarkRecord`, alongside reservation/quota snapshots, `PlacementPolicy`, and operation/audit records.

```
CustomerSeedRecord {                                # PLC-003/004/005 – first-placement author-
ity
  customer_realm_id      : string (PK)
  geography              : string (PK)
  production_region     : string
  cval_region           : string
  dr_region              : string | "NOT_OFFERED"
  products_covered      : list<string>
  decision_timestamp    : datetime
  policy_version        : string
  capacity_snapshot_ref : string
  exception_ref         : string | null
  approval_metadata     : ApprovalRecord
  input_mode            : "SPECIFIC_REGION" | "GEOGRAPHY_EXCEPTION"
}

SourceDestinationDRIndex {                          # DR-018 – reverse-of-seed, drives activa-
tion
  source_region          : string (PK)
  destination_region     : string (PK)
  customer_realm_id      : string (PK)
  standby_instance_set   : list<InstanceRef>
  sku_family             : string
  quantity_vcpu          : int
  activation_state       : STANDBY | ACTIVATING | ACTIVE | FAILBACK_PENDING
  last_updated           : datetime
  policy_version         : string
}

CVALEarmarkRecord {                                # PLC-010 – prevents DR/CVAL double-count
  customer_realm_id      : string (PK)
  region                 : string (PK)
  cval_capacity_vcpu     : int
  dr_earmarked_vcpu      : int                    # portion counted toward DR, not live CVAL
headroom
  co_located             : bool
  last_updated           : datetime
}
```

Supporting entities: `RegionalSnapshot` (freshness-stamped capacity/quota/usage/zone data), `Place-mentPolicy` (weights, buffers, thresholds, catalogue — versioned), `ReservationState` / `QuotaPoolState`, `OperationRecord` (saga + compensation), `AuditEvent` (append-only), `ActivationRecord` (DR-019, §9.3), `ZoneMappingRecord` (FC-06), `SharingRelationship`, `DRDistribu-tionPlan`.

T5 — Data model / ERD (incl. seed record, SourceDestinationDRIndex, CVAL earmark)

```

erDiagram
    CustomerSeedRecord ||--o{ SourceDestinationDRIndex : "reverse view"
    CustomerSeedRecord ||--o{ CVALEarmarkRecord : "co-location"
    CustomerSeedRecord }o--|| PlacementPolicy : "policy_version"
    RegionalSnapshot ||--o{ QuotaPoolState : "captures"
    QuotaPoolState ||--o{ ReservationState : "governs"
    SourceDestinationDRIndex ||--o{ ActivationRecord : "activates"
    DRDistributionPlan ||--o{ SourceDestinationDRIndex : "aggregates"
    OperationRecord ||--o{ AuditEvent : "emits"

    CustomerSeedRecord {
        string customer_realm_id PK
        string geography PK
        string production_region
        string cval_region
        string dr_region
        string input_mode
        string policy_version
    }

    SourceDestinationDRIndex {
        string source_region PK
        string destination_region PK
        string customer_realm_id PK
        int quantity_vcpu
        string activation_state
    }

    CVALEarmarkRecord {
        string customer_realm_id PK
        string region PK
        int cval_capacity_vcpu
        int dr_earmarked_vcpu
        bool co_located
    }

    QuotaPoolState {
        string region PK
        int pool_limit
        int prod_reserved_floor
        int dr_earmark_vcpu
        int allocatable_nonprod
    }

```

6.3 Freshness & versioning

Every snapshot carries `collected_at` ; the readiness gate compares `now() - collected_at` against `max_snapshot_age` (RDY-004). Policy and scope files are versioned and decision-traceable; the version that drove any decision is persisted with it (DAT-005, GOV-003). [Decided]

7. State Model & Concurrency (§29, NFR-002/007)

7.1 Engine mode machine (five states)

```
STEADY_STATE → DR_DECLARATION_PENDING → DR_EVENT_ACTIVE → FAILBACK_PENDING →  
STEADY_STATE  
└─ INCIDENT HOLD (safe degraded)
```

Steady-state reconciliation and auto-increase run **only** in `STEADY_STATE` ; emergency transfer is callable **only** in `DR_EVENT_ACTIVE` . Entering `DR_EVENT_ACTIVE` does not auto-authorise service-impacting CVAL action — each action retains its own policy gate. `INCIDENT_HOLD` is the safe degraded state entered when inputs cannot be trusted. [Decided]

7.2 Concurrency & reservation-of-intent

- **Optimistic concurrency:** every mutating operation reads an expected version and fails the write on mismatch, forcing a re-read (NFR-007). [Decided]
- **Reservation-of-intent:** placement writes an atomic hold before the seed record so two concurrent onboardings cannot double-commit the same capacity (NFR-002). [Decided]
- **Saga + compensation:** multi-step mutations use `OperationRecord` with a compensation chain for rollback; `VM_ImpactRecord` is immutable for any VM state change. [Decided]

T6 — Five-state engine machine

```
stateDiagram-v2
    [*] --> STEADY_STATE
    STEADY_STATE --> DR_DECLARATION_PENDING: DR declared (authorised)
    DR_DECLARATION_PENDING --> DR_EVENT_ACTIVE: activation approved
    DR_DECLARATION_PENDING --> STEADY_STATE: declaration withdrawn
    DR_EVENT_ACTIVE --> FAILBACK_PENDING: failback authorised
    FAILBACK_PENDING --> STEADY_STATE: failback complete (reversible)
    STEADY_STATE --> INCIDENT_HOLD: inputs untrusted / stale
    DR_EVENT_ACTIVE --> INCIDENT_HOLD: safety trip
    INCIDENT_HOLD --> STEADY_STATE: inputs restored
```

8. Placement Scoring & Forecasting (§28)

8.1 Input modes and pipeline

The default input is an **exact production region** which ACRME **validates** (it does not derive). A **geography** input is an **exception path** requiring approval + customer acknowledgement, after which the derived region becomes the fixed seed (ADR-001, PLC-001/002). Restricted regions never enter scoring — they route to the exception workflow.

T7 — Staged placement pipeline

```
flowchart LR
    In[Placement request] --> HC[Hard constraints HC-1..HC-10]
    HC -->|fail| Rej[Reason state / exception]
    HC -->|pass| Sel{Input mode}
    Sel -->|exact region default| Val[Validate & accept]
    Sel -->|geography exception| Score[Score Standard regions PS_Prod]
    Score --> Pick[argmax -> fixed seed]
    Val --> Seq[Sequential CVAL then DR selection]
    Pick --> Seq
    Seq --> Seed[Write seed + DR index]
```

T8 — Prod region input modes (exact vs geography exception)

```

flowchart TD
    R[Region input] --> Q{Exact region or geography?}
    Q -->|Exact Standard| V[Validate HC-1..HC-10 -> accept, no argmax]
    Q -->|Exact Restricted| X[Exception Deployment Workflow]
    Q -->|Geography| E{Exception approved + acknowledged?}
    E -->|No| B[POLICY_BLOCKED]
    E -->|Yes| D[argmax PS_Prod over Standard regions -> fixed seed]

```

8.2 Scoring formulas (weights $\alpha=0.30/\beta=0.20/\gamma=0.25/\delta=0.15/\epsilon=0.10$)

All five weights are tunable `PlacementPolicy` constants summing to 1.0; every component is clamped
`Clamp(x)=max(0,min(1,x))` before weighting. [Assumed]

```

PS_Prod(r)    = 0.30·Clamp(nonprod.effective_free / prod.quantity)
               + 0.20·Clamp(prod.quota_headroom / prod.quota_limit)
               + 0.25·Clamp(1 - prod_customer_count / total_customers)
               + 0.15·Clamp(dr.coverage_ratio)
               + 0.10·Clamp(az_count / 3)

PS_NonProd(r) = 0.30·Clamp(nonprod.effective_free / nonprod.quantity)
               + 0.20·Clamp(nonprod.quota_headroom / nonprod.quota_limit)
               + 0.25·Clamp(1 - nonprod_customer_count / total_customers)
               + 0.15·Clamp(nonprod.effective_free / nonprod.quantity)
               + 0.10·Clamp(az_count / 3)

PS_DR(r)      = 0.30·Clamp(dr.free_slots / dr.quantity)
               + 0.20·Clamp(dr.quota_headroom / dr.quota_limit)
               + 0.25·Clamp(1 - dr_customer_count / total_customers)
               + 0.15·min(1.0, dr.coverage_ratio / dr_ratio_target)
               + 0.10·Clamp(az_count / 3)

```

`PS_Prod` is dual-purpose (derive on exception; validate/audit otherwise). Every candidate score + policy version + snapshot ref is written to the `OperationRecord` for deterministic replay. A reviewer-recommended pilot `PS_NonProd` variant (removing the α/δ duplication) is recorded in the Calculation Logic Reference; the formula above is the approved design-of-record. Scoring runs in shadow/recommendation mode until empirically validated. [Decided]

8.3 Quota-pool arithmetic (single-pool, §26/QUA-004)

```

Pool_Limit(region) = Prod_CRG_qty·vCPU·(1+prod_growth_buffer)
                   + NonProd_CRG_qty·vCPU·(1+nonprod_growth_buffer)
                   + DR_Earmark_vCPU(region)
                   + emergency_transfer_headroom_vcpu

Prod_Reserved_Floor = Prod_Used_vCPU + Prod_Growth_Buffer_vCPU          # never alloc-
atable to NonProd
DR_Earmark_vCPU     = Destination_DR_Requirement(region)·vCPU          # max-not-sum,
never allocatable
Allocatable_NonProd = Pool_Limit - Prod_Reserved_Floor - DR_Earmark_vCPU - Non-
Prod_Used_vCPU
Pool_Headroom       = Pool_Limit - Pool_Used
Emergency_DR_Available = Pool_Headroom + reclaimable_NonProd_above_committed

```

NonProd allocation fail-safes when `Allocatable_NonProd` reaches zero. A dual-validation detector re-computes the DR earmark from authoritative assignment data; disagreement disables automatic NonProd expansion until reconciled (Scenario 9). `[Decided]`

8.4 Forecasting vs reconciliation floor

```
Target Reserved Capacity = Allocated VM Count + Configured Buffer          # CAP-003
continuous floor
Forecast_Quantity          = ceil(Forecast_Peak * (1+Growth_Buffer) + DR_Buffer) #
ADR-004 proactive growth
```

The reconciliation floor is enforced every cycle; the forecast is advisory (horizon $\in \{30,60,90\}$ days) and raises `ForecastApproachingQuotaLimit` with 14-day lead time at 80% of quota limit. Auto-increase target = max(floor, forecast). `[Decided]`

9. DR Sizing & Activation Algorithms (§31, App. A.6/A.7/A.8, App. D)

9.1 Max-not-sum sizing (DR-017, ADR-005)

```
Destination_DR_Requirement(d) = MAX over each non-concurrent source s protected by d
                               ( Workload_Portion(s -> d) )
DR_Capacity_Gap(d)            = max(0, Destination_DR_Requirement(d) - Us-
able_Destination_Capacity(d))
Overcommit_Ratio(d)           = SUM(Workload_Portion(s->d)) / MAX(Workload_Portion(s-
->d))
```

Single-failure assumption (DR-001): one production region fails at a time, so destination `d` needs standby for its **largest** protected source only; standby is shared/overcommitted across mutually exclusive events. `Usable_Destination_Capacity(d)` may include approved bootstrap headroom, available CRG reservations, releasable CVAL (after earmark check), and approved sharing/expansion. A per-scope **SUM override (C-11)** covers contractual simultaneous-failure protection with no code change. `[Decided]`

`DR_NOT_OFFERED` geographies are excluded from this sizing (DR-014, DEC-001). Sources in a geography flagged `DR_NOT_OFFERED` — the Middle East today — contribute **no** `Workload_Portion(s → d)` term, produce **no** `SourceDestinationDRIndex` entry, and reserve **no** `DR_Earmark_vCPU` at any destination. Their production is still sized and governed, but they carry no DR requirement, so they never enter `Destination_DR_Requirement`, `DR_Capacity_Gap`, or `Overcommit_Ratio`. Should Legal clear DEC-001 (config flip `dr_not_offered["Middle East"] = false`, no code change), the affected sources begin contributing their max-not-sum term at that point. `[Documented]`

T12 — Max-not-sum sizing (overcommit ratio)

```

flowchart TB
    S1[Source R1 -> R2: 120 cores]
    S3[Source R3 -> R2: 80 cores]
    S1 --> R2
    S3 --> R2
    R2[Destination R2 standby<br/>SUM = 200 cores over-provisions<br/>MAX = 120 cores<br/>sized<br/>saving = 80 cores 40%<br/>Overcommit_Ratio = 200/120 ≈ 1.67]:::d
    classDef d fill:#d4edda,stroke:#28a745;

```

9.2 Standby activation waves & staged acquisition (DR-006, DR-019)

On authorised declaration the DR Orchestrator reads the `SourceDestinationDRIndex` for the failed source and activates customers in business-priority waves (P0 platform → P-1 highest → P-N), transitioning each `associated` → `allocated`. Per wave, staged acquisition runs in order:

```

Stage 1: approved bootstrap / pre-staged headroom (zero Azure wait)
Stage 2: available reservation + quota already in destination
Stage 3: shut down / disassociate eligible CVAL (after DR-010 authorised trigger)
Stage 4: share/reassign reservations within supported region/zone boundaries
Stage 5: allocate pooled quota to DR subscriptions
Stage 6: request additional Azure quota/capacity where required
Stage 7: report unrecoverable capacity gaps

```

Stages 1-2 start immediately from the single pool's headroom, so P0/P-1 recovery does not wait on throttled Azure deallocation APIs. `[Decided]`

9.3 Activation state tracking & failback (DR-013)

```

ActivationRecord {
    customer_realm_id, source_region, destination_region,
    priority_wave : int,
    activation_state : PENDING | ACTIVATING | ACTIVE | FAILED | FAILBACK_PENDING,
    stage_reached : int (1..7), capacity_gap_vcpu : int,
    activated_at, approved_by, incident_id
}

```

Failback reverses `ACTIVE` → `FAILBACK_PENDING` → `STANDBY` in reverse priority order; seed record and DR index are preserved (no re-derivation). `[Decided]`

T11 — DR standby activation sequence (index lookup → waves → staged acquisition)

```
sequenceDiagram
    participant DRC as DR coordinator
    participant DRO as DR Orchestrator
    participant IDX as SourceDestinationDRIndex
    participant QPM as Quota-Pool Manager
    participant AZ as Azure
    DRC->>DRO: Declare source-region failure (authorised)
    DRO->>DRO: engine_mode = DR_EVENT_ACTIVE
    DRO->>IDX: Query standby sets for failed source
    IDX-->>DRO: Customers + priority waves
    loop each priority wave (P0 -> P-1 -> P-N)
        DRO->>QPM: Stage1-2 draw pool headroom + reservation
        QPM-->>DRO: Earmark released (no cross-group transfer)
        alt deficit remains
            DRO->>QPM: Stage3 CVAL release / Stage4 share / Stage5 pooled quota
            DRO->>AZ: Stage6 request additional capacity/quota
            AZ-->>DRO: Granted or deficit
        end
        DRO->>DRO: associated -> allocated (ActivationRecord)
        DRO->>DRC: Wave complete + audit checkpoint
    end
    DRC->>DRO: Failback authorised
    DRO->>DRO: ACTIVE -> FAILBACK_PENDING -> STANDBY (reverse order)
```

9.4 Tier escalation (§32) — see §13 baseline

Emergency capacity during `DR_EVENT_ACTIVE` escalates Tier 1 (direct expansion) → Tier 2 (quota-neutral transfer within the shared pool) → Tier 3 (destructive, dual-approval, **blocked in Phase 1**). In the single-pool model Tier 2 is intrinsically quota-neutral (draw and release hit the same pool). `[Decided]`

T10 — Three-tier emergency transfer escalation

```
flowchart TD
    Ev[DR event active] --> T1{Tier 1 headroom available?}
    T1 -->|Yes| A1[DirectExpansion - automated additive]
    T1 -->|No| T2{Tier 2 approved?}
    T2 -->|Yes| A2[QuotaNeutralTransfer - reduce NonProd, expand DR in same pool]
    T2 -->|No| T3{Tier 3 allowed?}
    T3 -->|Phase 1: BLOCKED| M[Manual escalation]
    T3 -->|Later phase: dual-approval| A3[DestructiveTransfer - changes VM associations]
```

T13 — Distributed DR reference model (§12A)

Reused reference diagram (ADR-005 §12A): `adr/diagrams/acrme_three_region_capacity_model.png` — each region concurrently hosts Prod + CVAL + DR standby tagged with its `src Rn` source; each destination is sized by max-not-sum against its largest single protected source.

10. Steady-State Capacity Lifecycle (§30)

The normative 10-step sequence runs only in `STEADY_STATE` :

1. Detect threshold crossing (utilisation $\geq$ auto-increase threshold).
2. Re-read current CR, quota, sharing, assignment state.
3. Create CapacityIncreaseRequest.
4. Calculate target = max(Allocated+Buffer floor, Forecast_Quantity).
5. Require operator approval (Phase 1).
6. Submit quota action only if validated as required.
7. Wait for confirmed quota state – no assumed propagation SLA.
8. Update CR quantity.
9. Confirm actual quantity.
10. Refresh snapshot and close the request.

Triggers: `dr_autoincrease_threshold=0.35` , `prod/nonprod_autoincrease_threshold=0.20` , debounce cooldown 30 min per (region+CRG type). Guarded scale-down never drops below `Allocated` , within a DR window, during maintenance exclusion, or against cost policy; reductions go to zero, never deletion (CAP-009/010). [Decided]

T9 — Steady-state 10-step capacity lifecycle

```
sequenceDiagram
    participant REC as State Reconciler
    participant OP as Operator
    participant QPM as Quota-Pool Manager
    participant AZ as Azure
    REC->>REC: 1. Detect threshold crossing
    REC->>AZ: 2. Re-read CR/quota/sharing/assignment
    REC->>REC: 3. Create CapacityIncreaseRequest
    REC->>REC: 4. Target = max(Allocated+Buffer, Forecast)
    REC->>OP: 5. Require approval (Phase 1)
    OP-->>REC: Approved
    REC->>QPM: 6. Submit quota action if required
    QPM->>AZ: 7. Await confirmed quota state
    AZ-->>QPM: Confirmed
    QPM->>AZ: 8. Update CR quantity
    AZ-->>QPM: 9. Confirm actual quantity
    REC->>REC: 10. Refresh snapshot + close request
```

11. API Architecture (§35, INT-001..007)

11.1 Contracts

POST /readiness	→ DeploymentReadinessResult	(query; idempotent)
in : { customer_realm_id, region (exact), sku_family, environment, product, intended_demand }		
out: { readiness_state, reasons[], operation_handle, snapshot_ref, policy_version }		
POST /placement	→ PlacementResult	(idempotent; writes seed + DR index)
POST /capacity/increase	→ OperationRecord	(operator-gated; idempotent)
POST /dr/declare	→ OperationRecord	(authorised; flips engine_mode)
POST /dr/activate-wave	→ ActivationRecord[]	(per priority wave)
POST /quota/allocate	→ OperationRecord	(single-pool allocation)
POST /sharing/activate	→ OperationRecord	(zone-aligned share)
GET /audit, GET /state	→ read-only	

11.2 Idempotency, versioning, error/readiness codes

- Every mutating call carries an **idempotency key** and an **expected-state version**; retried calls re-turn the original result, mismatched versions return a concurrency error (INT-004..006, NFR-007).
[Decided]
- Readiness states are the machine-readable contract to AEP:

READY | READY_WITH_RISK | QUOTA_DEFICIT | RESERVATION_DEFICIT |
CAPACITY_UNAVAILABLE | STALE_STATE | POLICY_BLOCKED | VALIDATION_REQUIRED

- AEP **polls** operation handles rather than assuming synchronous completion, and never treats provider quota as proof of consumer deployability (INT-007, POC-001). [Assumed]

11.3 Readiness gate logic (RDY-001)

READY requires ALL of:

1. target_region ∈ approved_region_catalogue (REG-001)
2. az_count ≥ 1 in target_region (CAP-011)
3. sku ∈ supported_skus(target_region)
4. reservation_policy known for (subscription, region, zone, sku)
5. reservation exists (if required by policy)
6. reserved_quantity ≥ requested_count OR approved_over_allocation active
7. consumer_subscription_quota ≥ requested_units (QUA-013 – POC-gated)
8. snapshot_age ≤ max_snapshot_age_seconds (RDY-004)
9. no blocking policy exception or hard-constraint violation

Quota_Deficit = max(0, Requested_Deployment_Units - Available_Quota)

Available_Quota = Assigned_Regional_VM_Family_Quota - Current_Regional_VM_Family_Usage

12. Security (§36, GOV-001..009)

- **RBAC & managed identity:** the control identity holds least-privilege, scoped roles per target subscription; no standing user credentials for mutations (GOV-001/002). [Decided]
- **Approval gates:** capacity/quota increase, DR declaration, CVAL release, Tier 2 transfer, seed change, and overrides are policy-gated with recorded approver identity (GOV-005..007).
[Decided]
- **Break-glass:** an audited emergency-elevation path with dual control and time-boxed elevated RBAC; every break-glass action is immutably logged (GOV-008). [Decided]
- **Secrets:** managed identity + platform secret store; no secrets in config or code. [Decided]
- **Audit:** append-only AuditEvent for every decision and mutation with before/after values and correlation IDs (GOV-004/009, DAT-005). [Decided]

13. Observability Implementation (§37, OBS-001..005)

Signal	Metric / alert	Requirement
Buffer deficit	<code>reservation_buffer_deficit</code> gauge; alert when reserved < <code>Allocated+Buffer</code>	OBS-001, CAP-003
Per-destination DR coverage	<code>dr_max_source_coverage(d)</code> ; alert when usable < <code>Destination_DR_Requirement(d)</code> (the MAX, not the SUM)	OBS-002, DR-017
DR capacity gap	<code>dr_capacity_gap(d)</code> gauge	OBS-002
Source↔destination mapping	dashboard of the <code>SourceDestinationDRIndex</code>	OBS-004
Overcommit ratio	<code>overcommit_ratio(d)</code> ; alert above safety ceiling	FIN-006..008, POC-011
Activation state	per-wave <code>ActivationRecord</code> progress	OBS-003, DR-019
Idle cost	idle reservation cost surfaced	OBS-005, FIN-001/002
Staleness	<code>snapshot_age</code> ; alert/refresh on breach	RDY-004, NFR-002
Forecast lead-time	<code>ForecastApproachingQuotaLimit</code> (14-day lead at 80%)	OBS-001, ADR-004

All metrics carry region/environment/scope dimensions and policy version for replay. [Decided]

14. NFRs & Resilience (§38, NFR-001..010)

- **Throttling resilience:** adaptive throttle manager initialises with ARM baselines (250 reads/5 min, 1,200 writes/hour per subscription) and adapts to observed `429/Retry-After` ; reconciliation cadence and estate size are bounded accordingly (§14 calc). [Documented]
- **Estate sizing:** `CRG_total = R·Eclass·Z·SKUset·IsolationFactor` ; `Requests_per_cycle` summed across read classes / 300 s cycle to stay within budget. [Derived]
- **Degraded mode:** on untrusted/stale inputs the engine enters `INCIDENT_HOLD` / returns `STALE_STATE` and fails safe rather than driving placement from stale data (NFR-002). [Decided]
- **Simulation / dry-run:** placement, reconciliation, and **DR failover** can run in dry-run/simulation, producing the would-be decisions and readiness states without mutating Azure (NFR-006). [Decided]

- **Idempotency & concurrency:** all mutations idempotent + optimistic concurrency (NFR-007). Determinism/replay from `OperationRecord`. [Decided]

15. Integration (§33, §6 review)

- **AKS / VMSS:** reservations back node pools / scale sets; ACRME readiness gates their capacity before scale-out; associations are tracked for reconciliation. [Decided]
- **AEP:** the sole provisioning consumer of readiness; interacts only through the idempotent readiness/placement contract (§11) and polls operation handles. [Decided]
- **Azure control planes:** Compute (CRGs/reservations), Reservations, and **groupQuotas** (single-pool). Single-pool enforcement depends on `groupQuotas` / `groupType` maturity (DEP-001). [Documented]

16. POC & Validation Dependencies (§42)

POC	Question	Gates	Status
POC-001	Does provider quota under a shared reservation guarantee consumer deployability?	Readiness gate step 7; INT-007	Top unknown
POC-006	Distributed DR topology behaves as modelled	ADR-005 §12A; DR-016	Required
POC-007	Lean bootstrap sizing is sufficient to initiate recovery	DR-007 bootstrap targets	Required
POC-011	max-not-sum over-commit is safe at platform scale	DR-017 sizing; over-commit safety ceiling; FIN-006..008	Required before production dependency
POC-005	VM state semantics (associated/allocated/deallocated)	DR-019 activation staging	Dependency
POC-031/032	Tier 2 quota-neutral transfer within a group/pool	§13 baseline Tier 2	Required

Production reliance on max-not-sum DR sizing and single-pool consumer-quota behaviour is **gated** on POC-011 and POC-001 respectively; until then, readiness validates consumer quota explicitly and DR sizing carries the SUM-override safety option. [Decided]

17. Traceability — Requirement/Deviation → Component/Algorithm/Entity

Requirement group (v2.2 IDs)	Component(s)	Algorithm / formula	Entity / diagram
REG-001..003	Config/Scope-File, Placement Engine	catalogue validation; input modes	PlacementPolicy ; T8
ENV-001..007	Placement Engine, DR Orchestrator	env separation; role flip	CustomerSeedRecord ; T7
CAP-001..019	Inventory Collector, State Reconciler	Allocated+Buffer floor; zero-not-delete; over-alloc	ReservationState ; T9, F5
QUA-001..014	Quota-Pool Manager	§8.3 pool arithmetic; earmarks; quota-as-governor	QuotaPoolState ; T4
RDY-001..004	Readiness API	§11.3 gate logic; staleness; quota deficit	readiness states; T6
PLC-001..010	Placement & Scoring Engine	PS_Prod/NonProd/DR ; seed reuse; co-location guard	CustomerSeedRecord , CVALEarmarkRecord ; T7, T8
DR-001..019	DR Orchestrator, Emergency Transfer	max-not-sum; gap; overcommit; activation waves; staged acquisition; tiers	SourceDestination-DRIndex , ActivationRecord ; T11, T12, T13, T10
FIN-001..008	Observability Emitter, Forecast Service	idle cost; overcommit ratio; cost-before-expansion	metrics; T12
INT-001..007	Readiness API	idempotency keys; expected-version; polling	OperationRecord ; T1
DAT-001..006	State Store, Config Service	freshness metadata; versioning; audit	all entities; T5
OBS-001..005	Observability Emitter	max-source coverage; source↔dest map; activation progress	dashboards; §13
GOV-001..009	Governance & Audit	RBAC; approval gates; break-glass; immutable audit	AuditEvent ; §12

Requirement group (v2.2 IDs)	Component(s)	Algorithm / formula	Entity / diagram
NFR-001..010	(cross-cutting)	throttle resilience; degraded mode; idempotency; simulation	OperationRecord ; §14
OPS-001..005	(runbooks)	DR declaration/activation/failback; quota allocation; sharing; override	ActivationRecord ; §10, §16
POC-001..011	(gating)	consumer quota; DR topology; bootstrap; overcommit safety	§16
DEC-001..003 / DEP-001	Config/Scope-File, Quota-Pool Manager	Middle East DR policy — current position DR_NOT_OFFERED , pending legal review (production allowed, no DR, Switzerland North path inactive until config flip); failback duration; geo-exception approver; groupQuotas maturity	PlacementPolicy ; §2.2, §5.3

Every Baseline v2.2 requirement group resolves to at least one component, algorithm, and entity above. Functional-level traceability is in the FDD §9.

18. Class Diagram (updated with new entities)

T14 — Domain class model (updated: +CustomerSeedRecord, +SourceDestinationDRIndex, +CVALEarmarkRecord; single-pool QuotaPoolState ; readiness enum)

```

classDiagram
    class CustomerSeedRecord {
        +string customer_realm_id
        +string geography
        +string production_region
        +string cval_region
        +string dr_region
        +string input_mode
        +string policy_version
    }
    class SourceDestinationDRIndex {
        +string source_region
        +string destination_region
        +string customer_realm_id
        +int quantity_vcpu
        +ActivationState activation_state
    }
    class CVALEarmarkRecord {
        +int cval_capacity_vcpu
        +int dr_earmarked_vcpu
        +bool co_located
    }
    class QuotaPoolState {
        +int pool_limit
        +int prod_reserved_floor
        +int dr_earmark_vcpu
        +int allocatable_nonprod
        +int pool_headroom
    }
    class PlacementPolicy {
        +float alpha_beta_gamma_delta_epsilon
        +float prod_growth_buffer
        +float nonprod_growth_buffer
        +string version
    }
    class ActivationRecord {
        +int priority_wave
        +int stage_reached
        +int capacity_gap_vcpu
        +ActivationState state
    }
    class ReservationState {
        +int reserved_quantity
        +int allocated_count
    }
    class OperationRecord {
        +string operation_handle
        +string expected_version
    }
    CustomerSeedRecord "1" --> "0..*" SourceDestinationDRIndex : reverse view
    CustomerSeedRecord "1" --> "0..*" CVALEarmarkRecord
    CustomerSeedRecord --> PlacementPolicy : policy_version
    SourceDestinationDRIndex "1" --> "0..*" ActivationRecord
    QuotaPoolState "1" --> "0..*" ReservationState
    OperationRecord --> ReservationState : mutates

```

Note: the earlier UML summary’s “two quota groups per region” is superseded by the **single governed QuotaPoolState** with logical earmarks (exception two-group topology only). [Decided]

Document Status: Draft for review · **Next:** ratify alongside the FDD; supersede the Production Readiness Review (archived). Production reliance on max-not-sum DR and single-pool consumer quota is gated on POC-011 / POC-001.